

Models of the motor system

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Part 1

- Illusions and Perception
 - Perception is a constructive process
 - Transform signals into internal models
 - Illusions reveal underlying computations

Sensation and perception

- Last class we studied sensation
 - How does the information get into the system
- Today we'll talk about perception
 - How does the information get processed
 - Into what we experience
 - Into a model of what is out there

Better hitters see a bigger ball

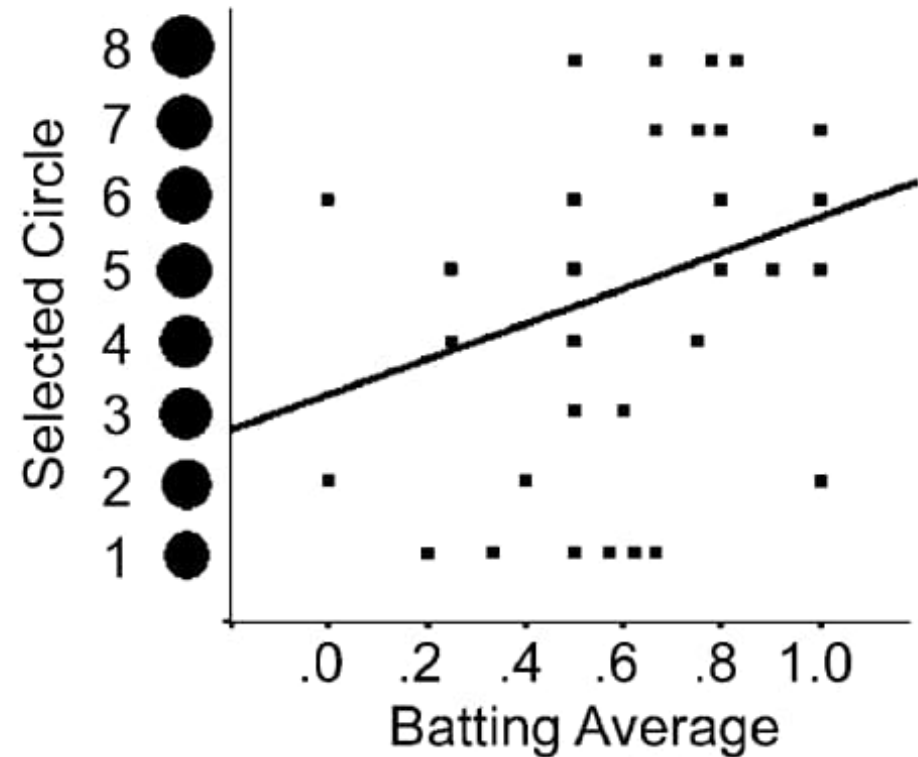
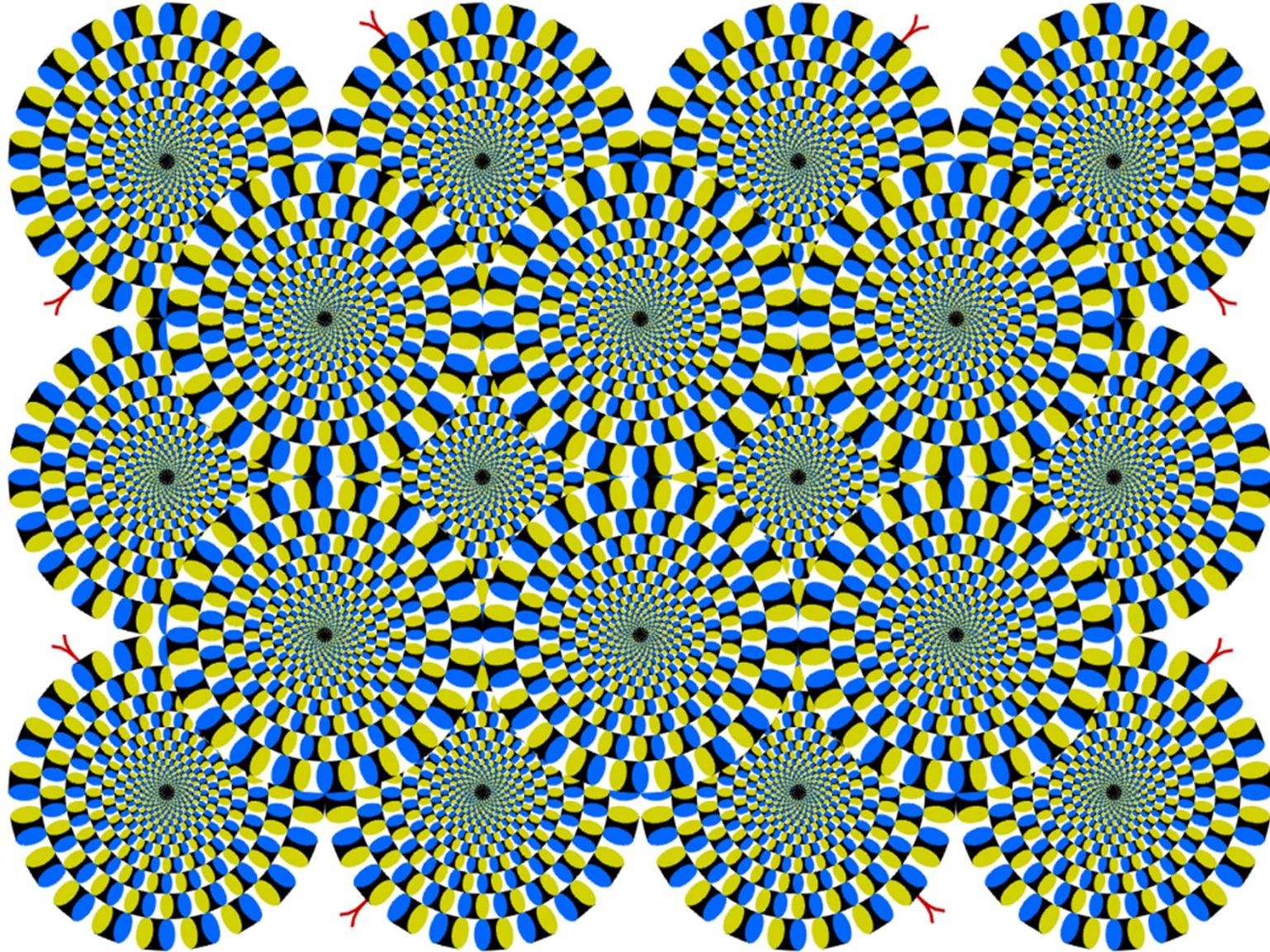
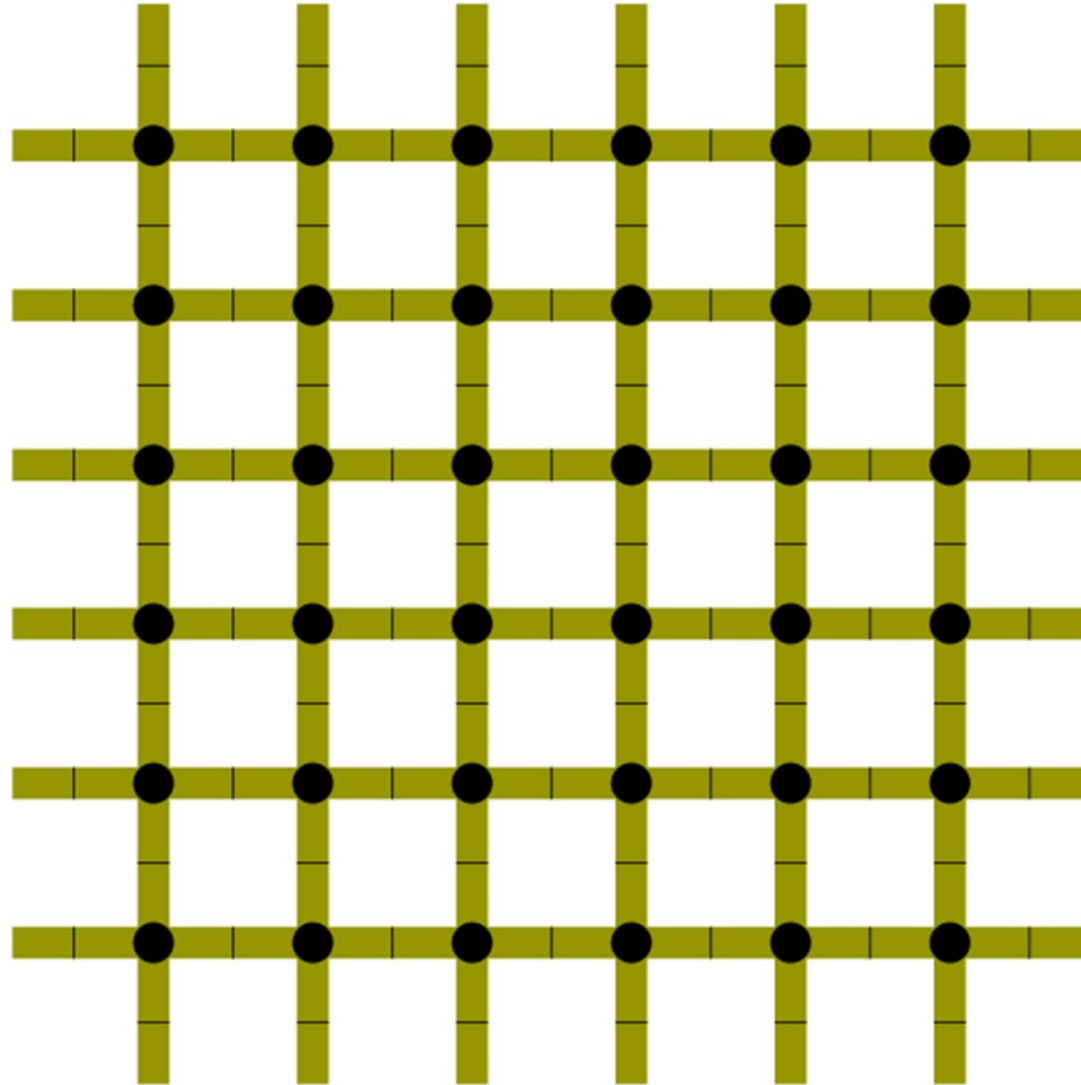


Fig. 1. Apparent ball size as a function of batting average. Each square represents 1 or more participants' data. The circles on the y-axis are drawn to preserve relative size. The solid line is the correlation between batting average during one or two softball games and the circle selected as best matching the size of the softball.

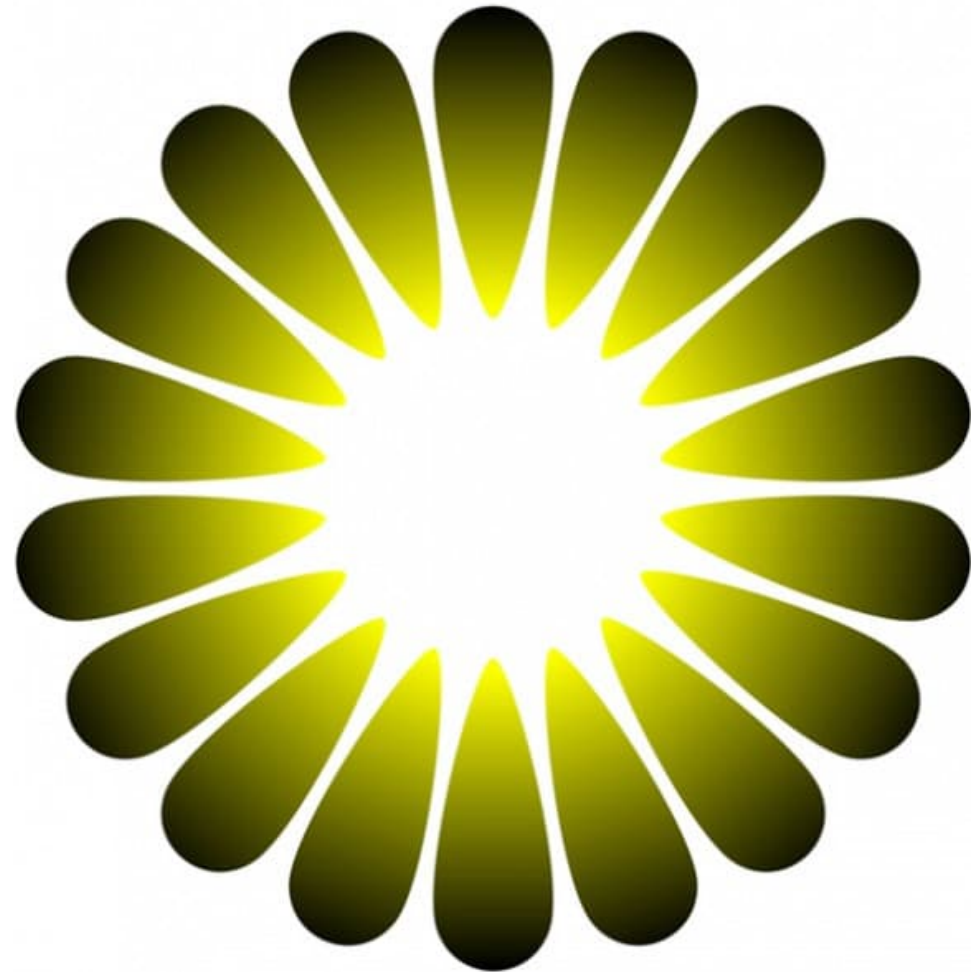
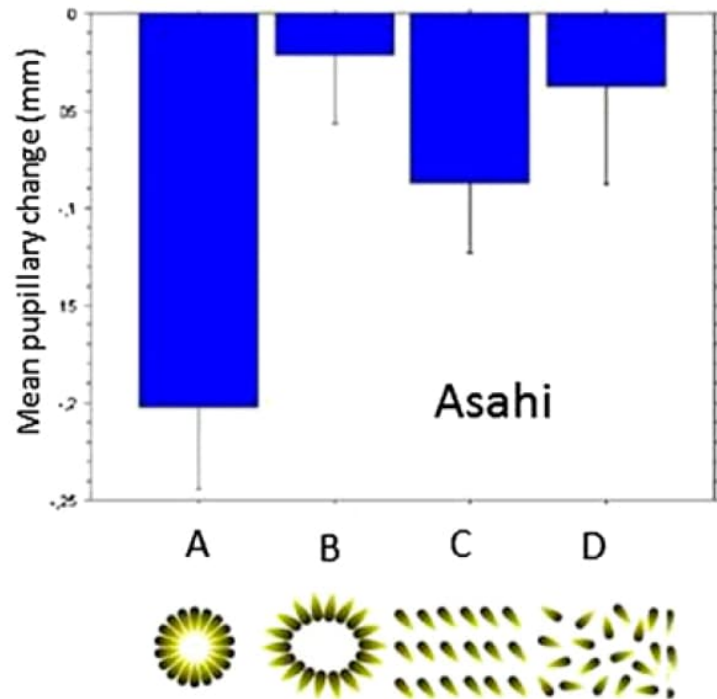
Study perceptions through illusions



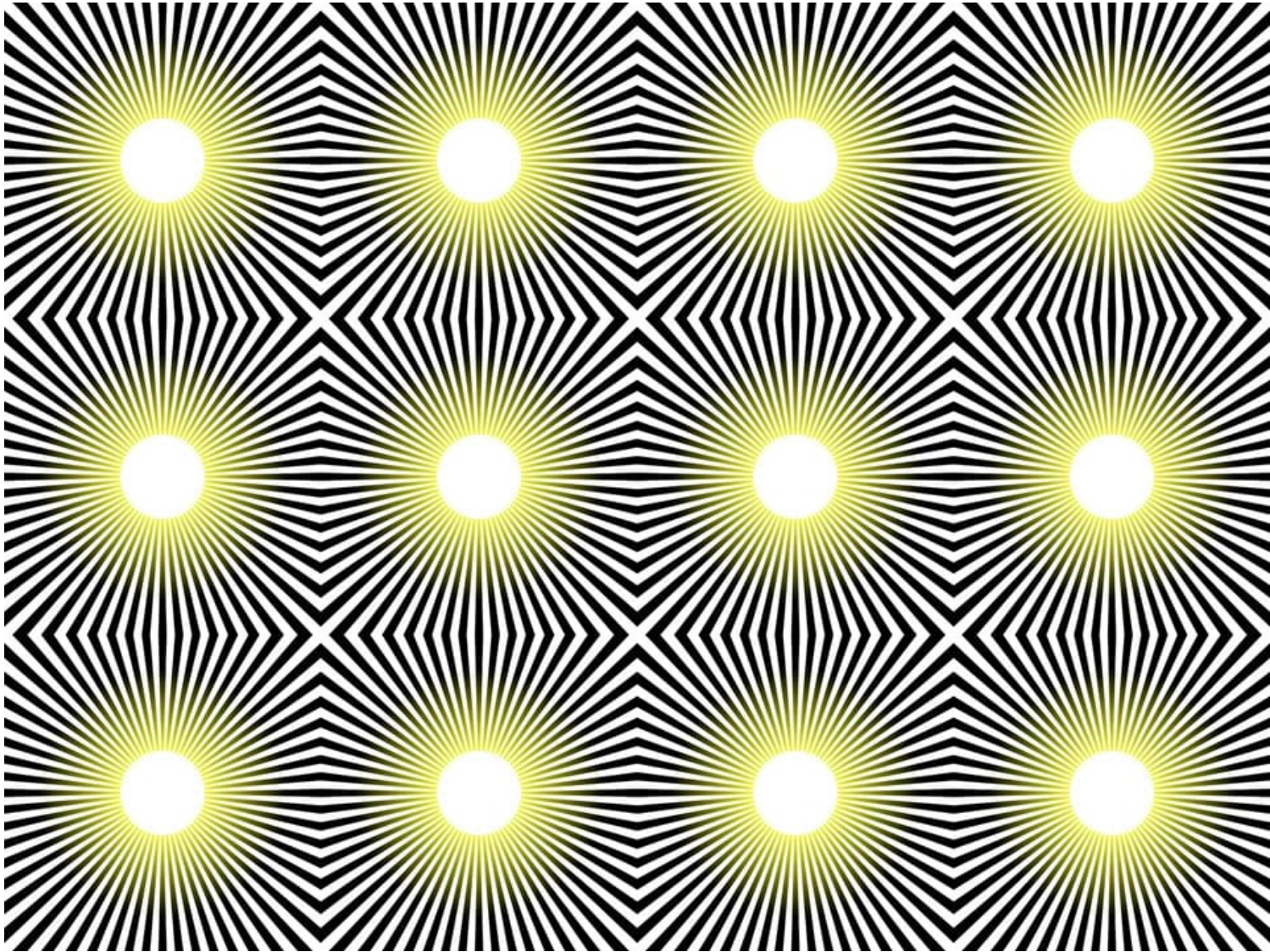
Retinal contrast calculations
create this illusion



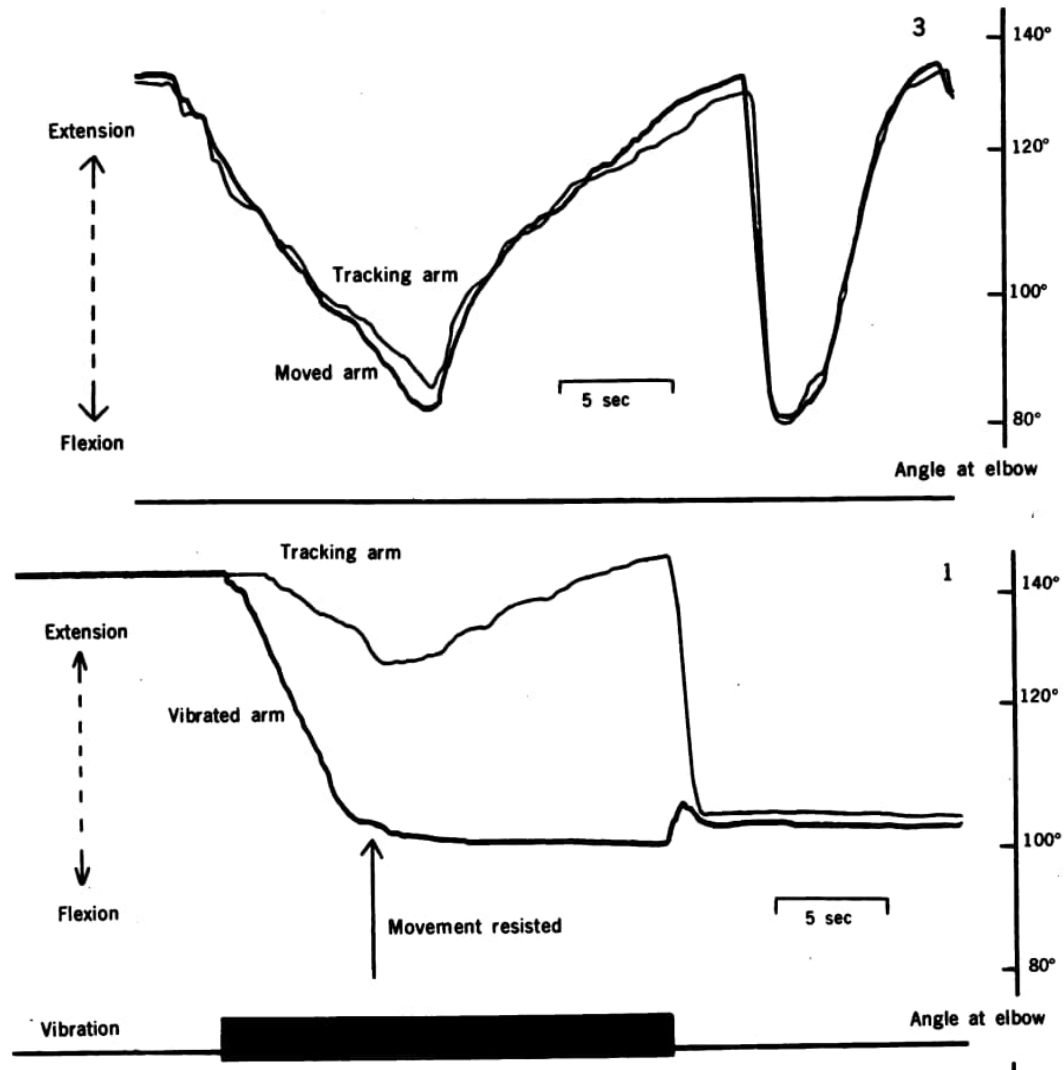
This illusion affects pupil size



Stronger version of the effect



Proprioceptive illusion



Vibration of the flexor muscle tendon causes the arm to flex and a sense that the arm is being extended.

Goodwin *Nature* 1978 175:1382

Illusions teach about sensory processing

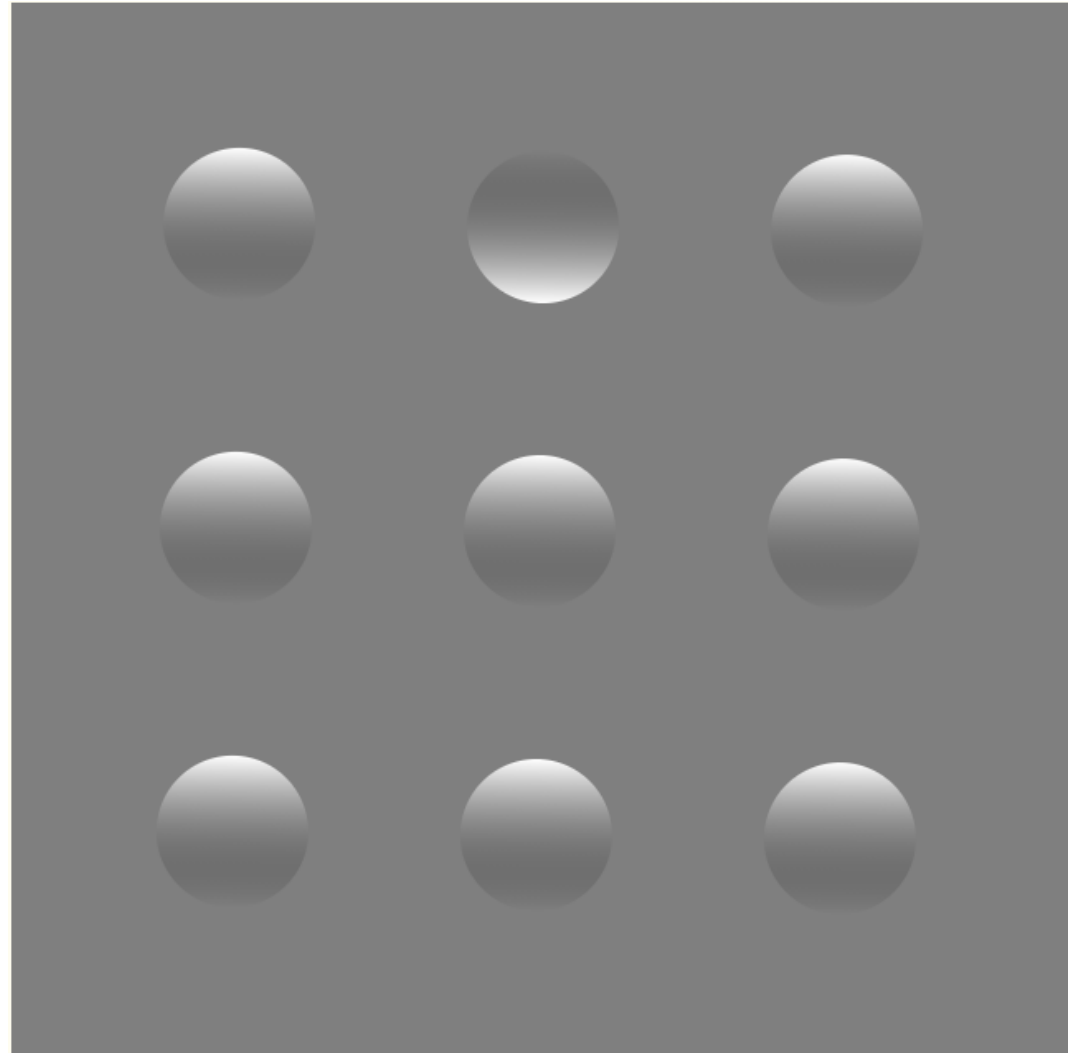
- Vibrated tendon illusion
 - Excites muscle tendon organ
 - Signal indicates increased muscle tension
 - Increase muscle activation
 - Arm moves
 - But feeling increased tension continues
 - Feel like the arm is extending

Part 2

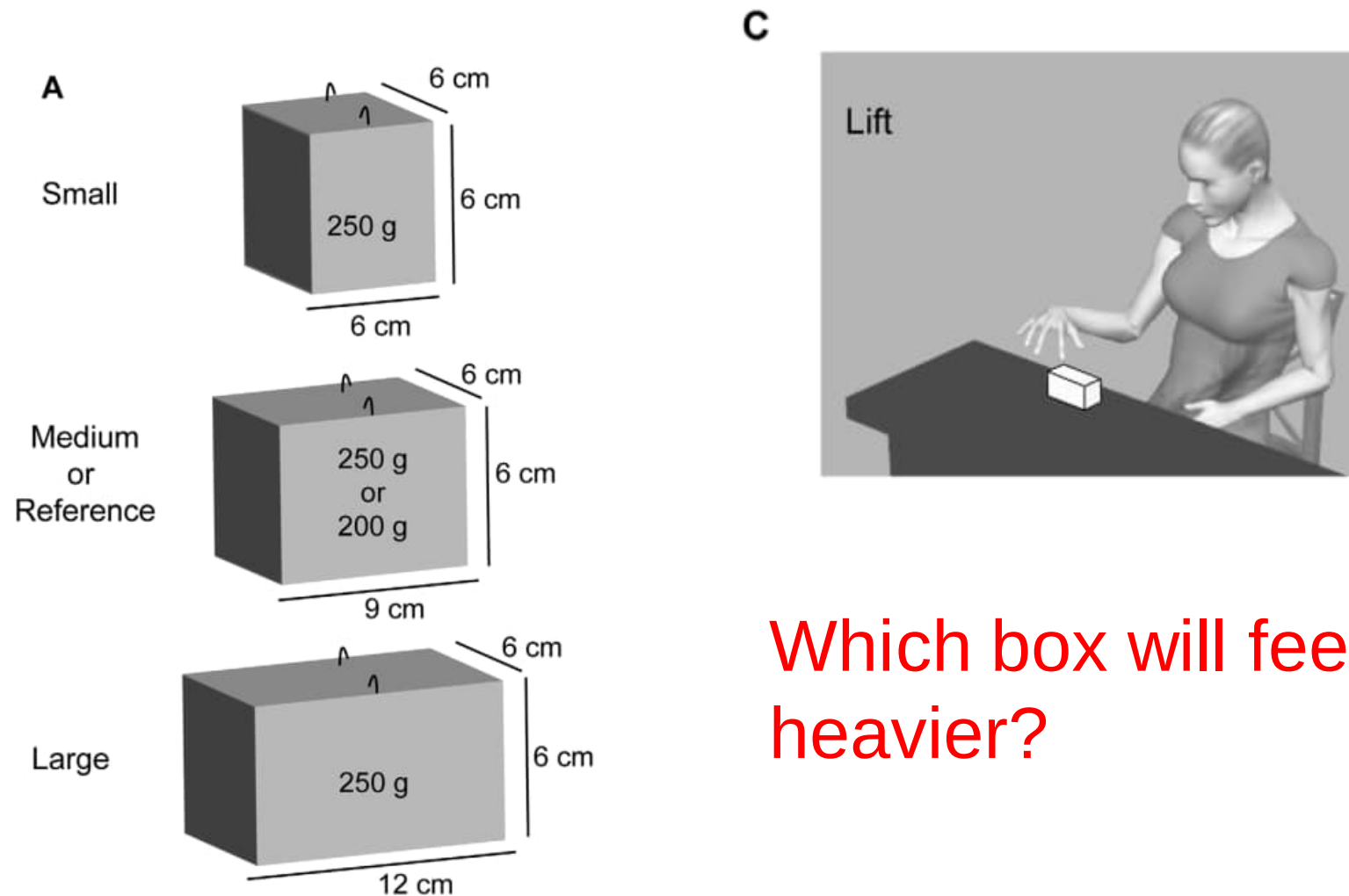
- Expectations and the Motor System
 - Perceptual illusions reflect prior expectations
 - Motor pathways can remain accurate despite perceptual errors

Illusions can reflect expectations

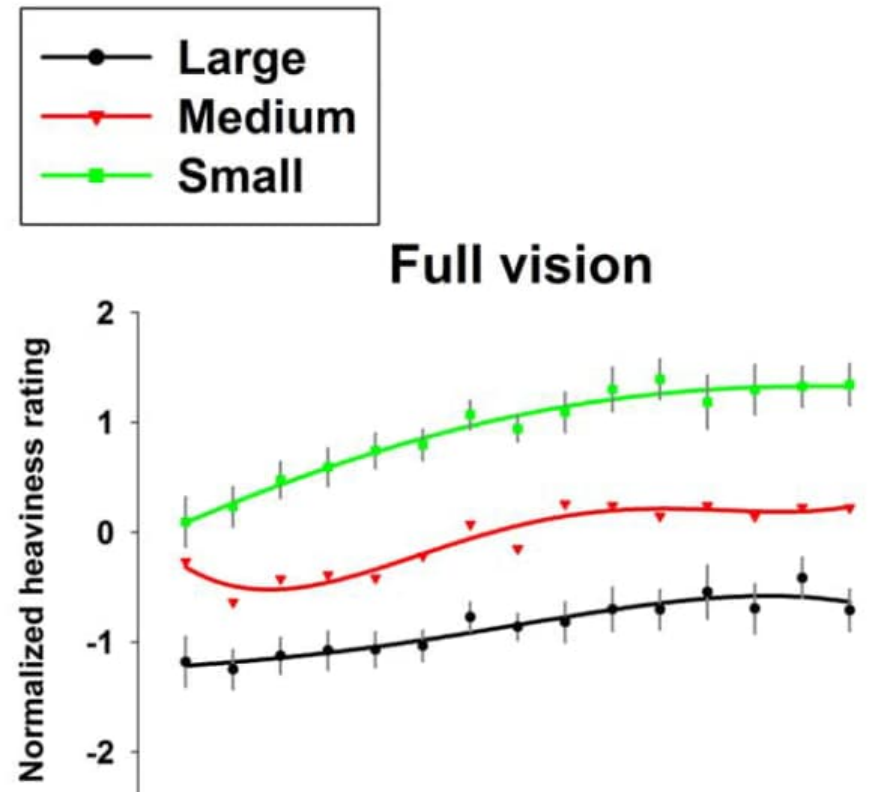
- Light comes from above



Size / weight illusion

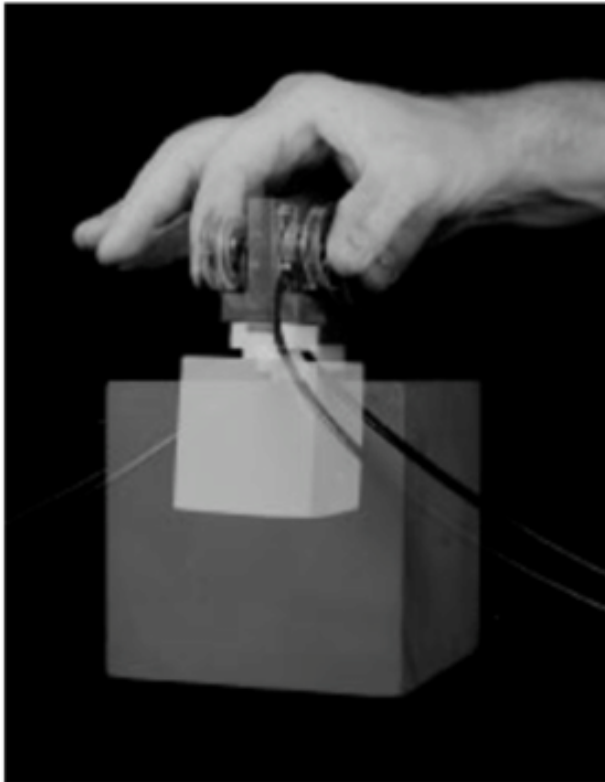


First reported in 1889

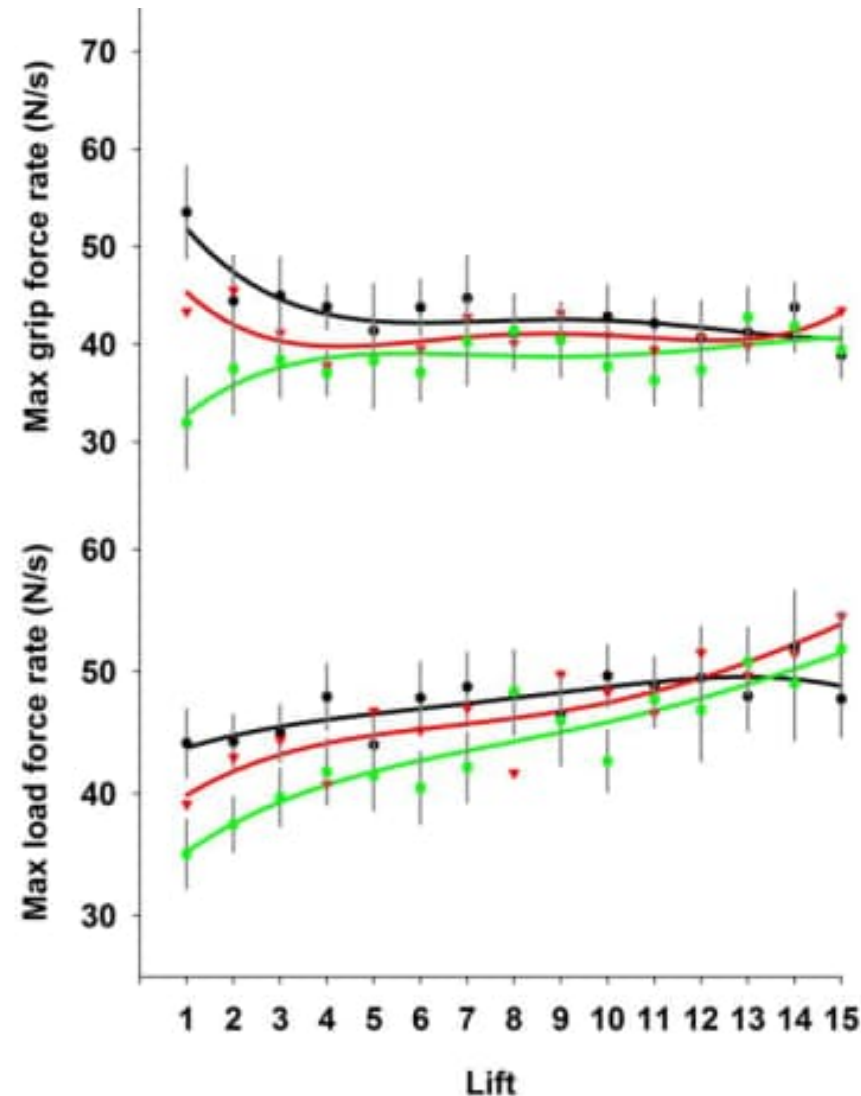
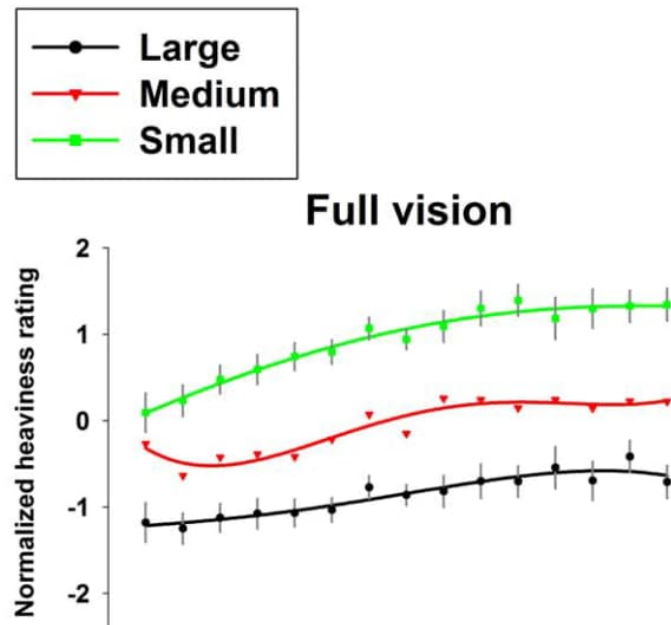


Is the motor system also fooled?

Device for testing the SWI when two objects of different dimension and same weight are lifted in alternation. This device also allows measuring the grip forces exerted in each case [Flanagan & Beltzner, 2003].



The motor system is not fooled



Explanations of the size weight illusion

- Greater motor output makes heavy object feel lighter
 - Illusion persists after motor output adapts
- Expected weight affects weight perception
 - Size-material illusion
 - Expectation can be reversed with enough practice

Part 3

- Multisensory illusions
 - The brain integrates information from multiple sense
 - Flexible remapping across modalities

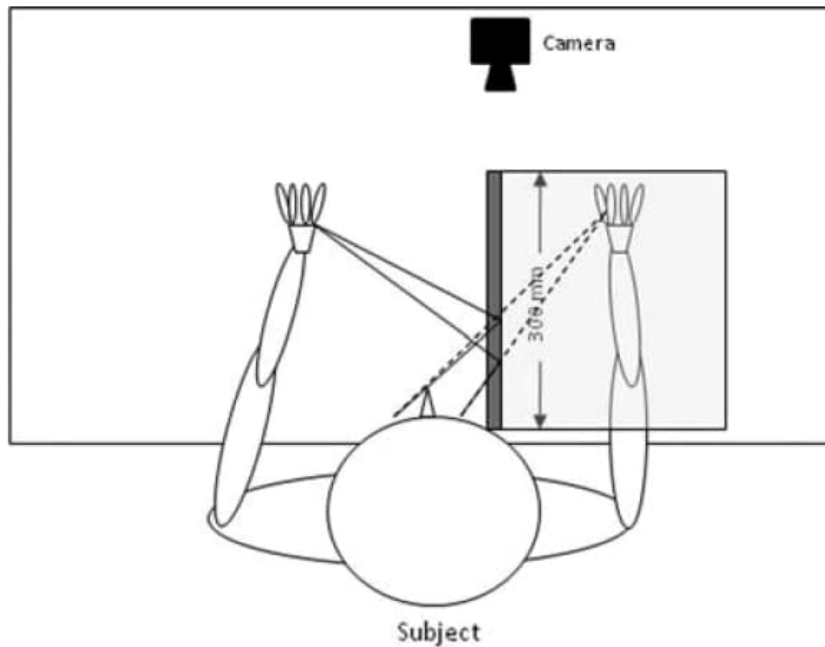
Illusions of multiple senses

- The McGurk effect

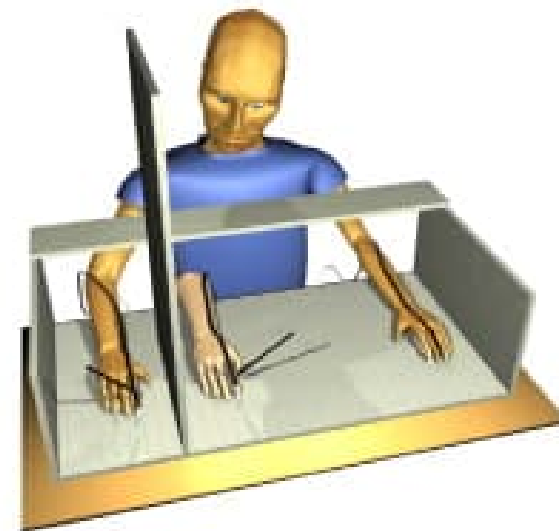


The mirror box and the rubber hand

- Two illusions that give us the feeling we are seeing both hands



(A)



The mirror box illusion



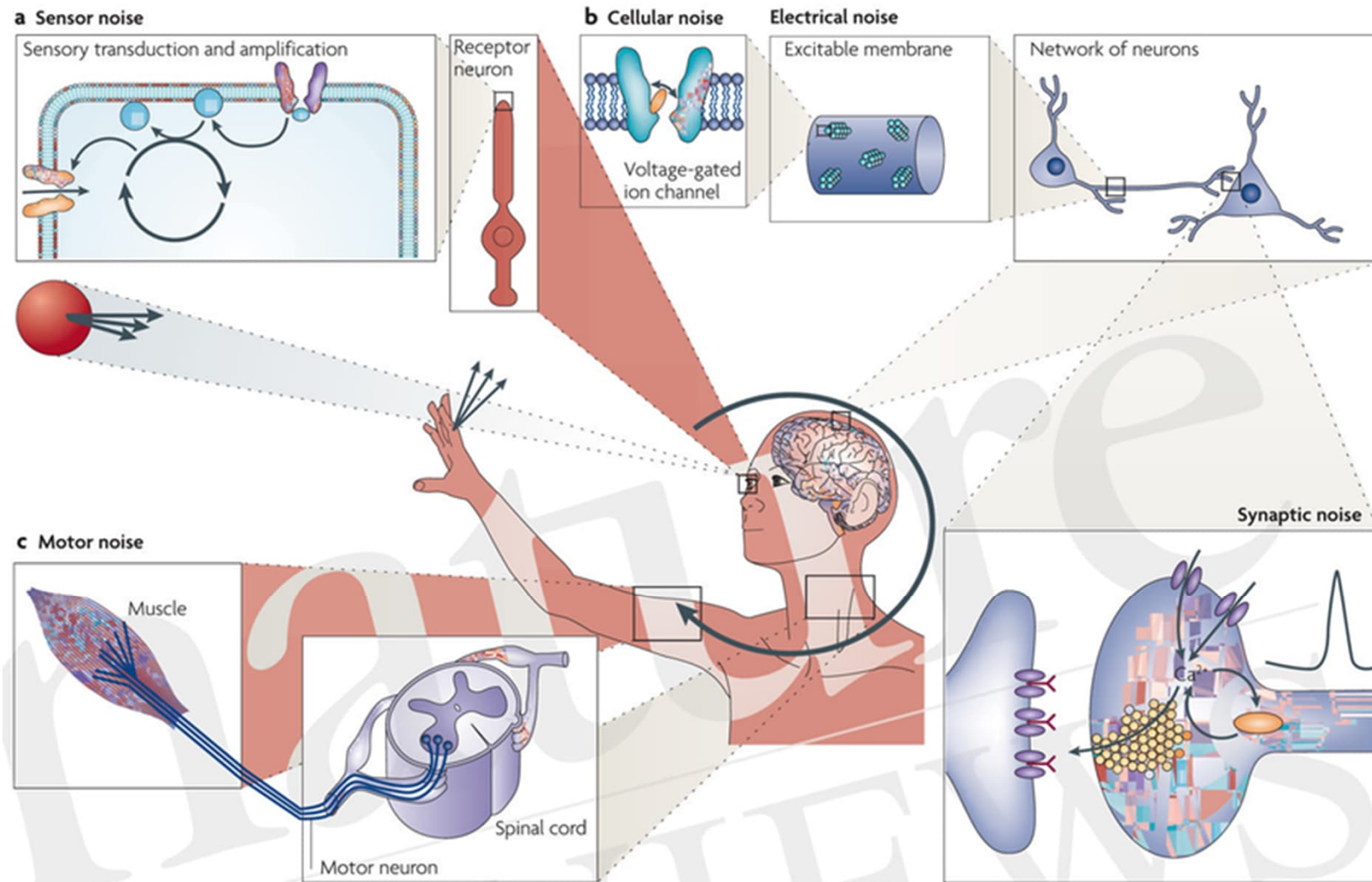
The rubber hand illusion



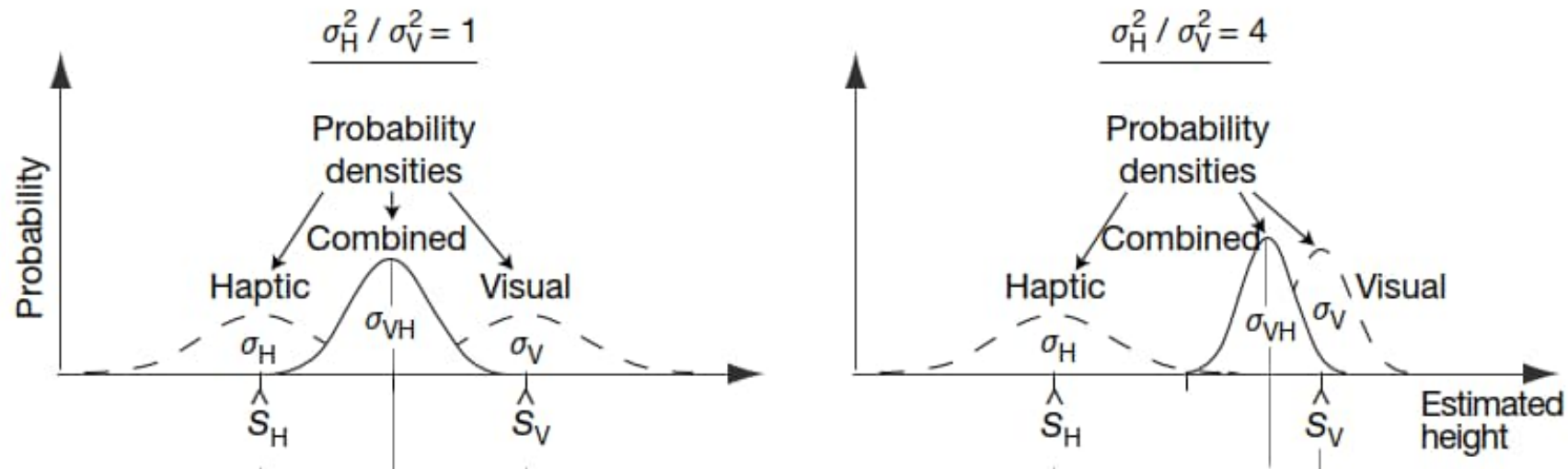
Part 4

- Bayesian estimation in perception
 - The optimal combination of multiple sources of information
 - Each sense weighted by its reliability

Senses combine in the face of noise



Noise levels affect optimal combination



- We put more weight on the sensory input with less noise
- Optimal combination is done using Bayesian estimation

Bayesian estimation explains a lot

- The role of expectations
- Adaptation with repeated exposure
- Weighting of different estimates

Bayes rule: optimally combining knowledge with experience



James Bernoulli (1713) asked:
Do rules of probability apply in inductive logic.



Dana Ballard (1999) asked:
Do inductive rules of probability govern what we perceive.

What is a joint distribution?

Eye Color	Hair Color				Marginal (Eye Color)
	Black	Brunette	Red	Blond	
Brown	.11	.20	.04	.01	.37
Blue	.03	.14	.03	.16	.36
Hazel	.03	.09	.02	.02	.16
Green	.01	.05	.02	.03	.11
Marginal (Hair Color)	.18	.48	.12	.21	1.0

$$p(e = \text{black}, h = \text{brown}) = 0.11$$

$$p(e, h) = p(h, e)$$

$$p(e) = \sum_h p(h, e)$$

$$p(h) = \sum_e p(h, e)$$

What is a conditional probability?

Eye Color	Hair Color				Marginal (Eye Color)
	Black	Brunette	Red	Blond	
Brown	.11	.20	.04	.01	.37
Blue	.03	.14	.03	.16	.36
Hazel	.03	.09	.02	.02	.16
Green	.01	.05	.02	.03	.11
Marginal (Hair Color)	.18	.48	.12	.21	1.0

Eye Color	Hair Color				Marginal (Eye Color)
	Black	Brunette	Red	Blond	
Blue	$\frac{.03}{.36}$ = .08	$\frac{.14}{.36}$ = .39	$\frac{.03}{.36}$ = .08	$\frac{.16}{.36}$ = .45	$\frac{.36}{.36}$ = 1.0

What is a conditional probability?

Eye Color	Hair Color				Marginal (Eye Color)
	Black	Brunette	Red	Blond	
Blue	.03/.36 = .08	.14/.36 = .39	.03/.36 = .08	.16/.36 = .45	.36/.36 = 1.0

$$p(h|e) = \frac{p(h,e)}{p(e)} = \frac{p(h,e)}{\sum_{h^*} p(h^*,e)}$$

Baye's rule

$$p(c|r) = \frac{p(c,r)}{p(r)}$$

$$p(r|c) = \frac{p(c,r)}{p(c)}$$

$$p(c|r)p(r) = p(c,r)$$

$$p(r|c)p(c) = p(c,r)$$

$$p(c|r)p(r) = p(r|c)p(c)$$

$$p(c|r) = \frac{p(r|c)p(c)}{p(r)} = \frac{p(r|c)p(c)}{\sum_{c^*} p(r|c^*)p(c^*)}$$

Baye's rule from a table

Row	Column			Marginal
	...	c	...	
$\vdots$		$\vdots$		
r	...	$p(r, c)$ $= p(r c) p(c)$	...	$p(r)$ $= \sum_{c^*} p(r c^*) p(c^*)$
$\vdots$		$\vdots$		
Marginal		$p(c)$		

By definition

$$p(r|c) \equiv \frac{p(r, c)}{p(c)}$$

Bayes rule

$$p(c|r) = \frac{p(r, c)}{p(r)} = \frac{p(r|c) p(c)}{\sum_{c^*} p(r|c^*) p(c^*)}$$

Applied to Estimation

$$p(c|r) = \frac{p(r|c)p(c)}{\sum_{c^*} p(r|c^*)p(c^*)}$$


$$p(\text{reality}|\text{sense data}) = \frac{p(\text{sense data}|\text{reality}) p(\text{reality})}{\sum_{\text{parameters}} p(\text{sense data}|\text{reality}) p(\text{reality})}$$

Data	Model Parameter			Marginal
	...	θ value	...	
$\vdots$		$\vdots$		$\vdots$
D value	...	$p(D, \theta)$ $= p(D \theta) p(\theta)$	...	$p(D)$ $= \sum_{\theta^*} p(D \theta^*) p(\theta^*)$
$\vdots$		$\vdots$		$\vdots$
Marginal	...	$p(\theta)$	...	

Applied to new information

$$p(c|r) = \frac{p(r|c)p(c)}{\sum_{c^*} p(r|c^*)p(c^*)}$$


$$p(\theta|D) = \frac{p(D|\theta)p(\theta)}{\sum_{\text{parameters}} p(D|\theta)p(\theta)} = \frac{p(D|\theta)p(\theta)}{p(D)}$$

- Likelihood: $p(D|\theta)$ ← How likely is the data for different possible realities?
- Prior: $p(\theta)$ ← What did we believe before we saw the data?
- Evidence: $p(D) = \sum p(D|\theta)p(\theta)$
- Posterior: $p(\theta|D)$ ← What do we believe now that we did see that data?

Part 5

- Bayes' rule: examples

VR Plank Experience



Prior information

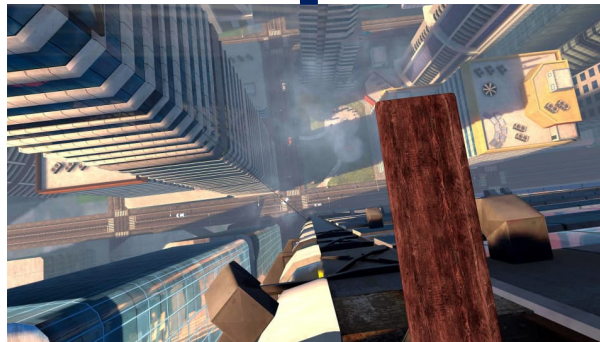
- You KNOW you are on the floor
- You SEE you are on a plank in the air



$p(\text{in room})$



$p(\text{in room} \mid \text{what I'm seeing})$



$p(\text{what I'm seeing} \mid \text{in room})$

Working through the calculations

$$\begin{aligned} p(\text{up high} \mid \text{what I'm seeing}) &= \frac{p(\text{what I'm seeing} \mid \text{up high}) p(\text{up high})}{p(\text{what I'm seeing})} \\ &= \frac{p(\text{what I'm seeing} \mid \text{up high}) p(\text{up high})}{p(\text{what I'm seeing} \mid \text{up high}) p(\text{up high}) + p(\text{what I'm seeing} \mid \text{in room}) p(\text{in room})} \\ &= \frac{0.9999 \cdot 0.001}{0.9999 \cdot 0.001 + 0.0001 \cdot 0.999} \end{aligned}$$

$$p(\text{up high} \mid \text{what I'm seeing}) = 0.90$$

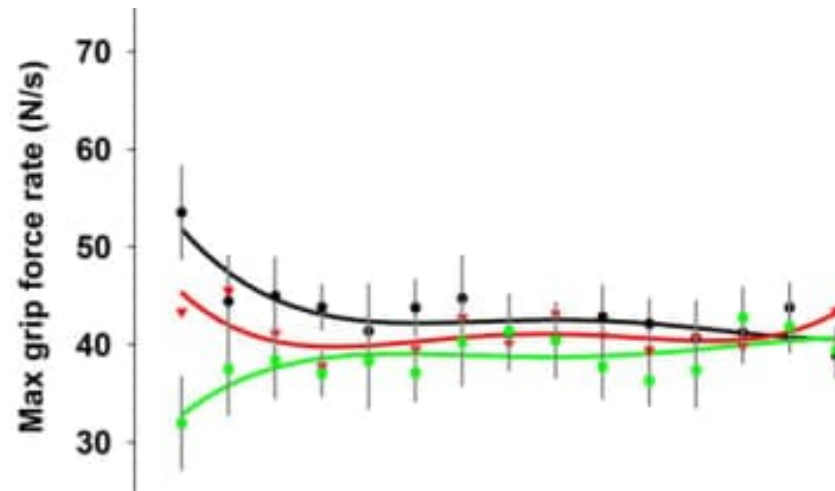
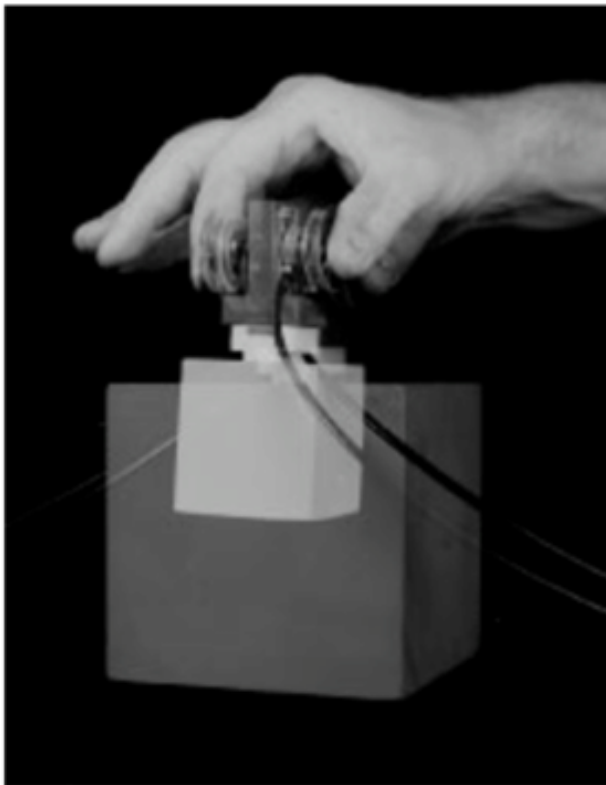
The importance of multi-sensory integration

$$p(\text{up high} \mid \text{what I'm seeing}) = \frac{p(\text{what I'm seeing} \mid \text{up high}) p(\text{up high})}{p(\text{what I'm seeing})}$$

- Input from multiple senses combines to make very strong likelihoods
- The data from our senses ultimately overcomes our priors
 - Our sense create our reality

Bayesian updating explains adaptation

- Initial grip force reveals expected weight
- Change in grip force reflects updating



Simple example problem

- I give you a (possibly biased) coin
- Flip it N times, count z heads
- What can you tell me about this bias of the coin?

The likelihood function

- Probability of data given the parameter

$$p(y|\theta) \quad \text{This is a probability} \quad \int dy \, p(y|\theta) = 1$$

For a coin with probability θ , the Bernoulli distribution

$$p(y|\theta) = \theta^y (1-\theta)^{(1-y)}$$

$$p(1|\theta) + p(0|\theta) = \theta + (1-\theta) = 1$$

$$p(\{y_i\}|\theta) = \theta^z (1-\theta)^{N-z}$$

z : Number of 1s

N : Number of attempts

Coin flip experiment

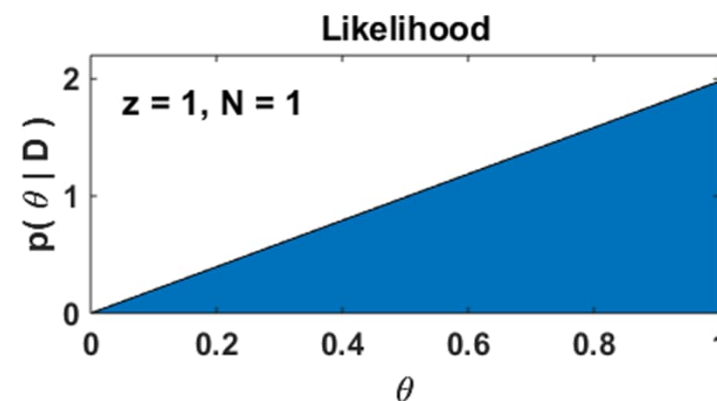
Heads: $y = 1$

$$p(y = 1 | \theta) = \theta$$

Tails: $y = 0$

$$p(y = 0 | \theta) = 1 - \theta$$

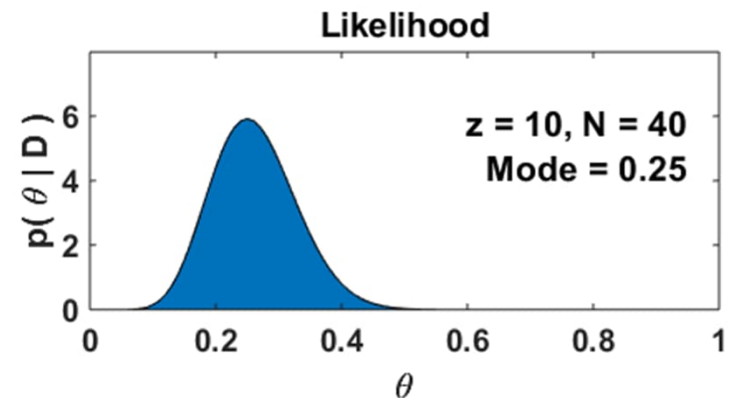
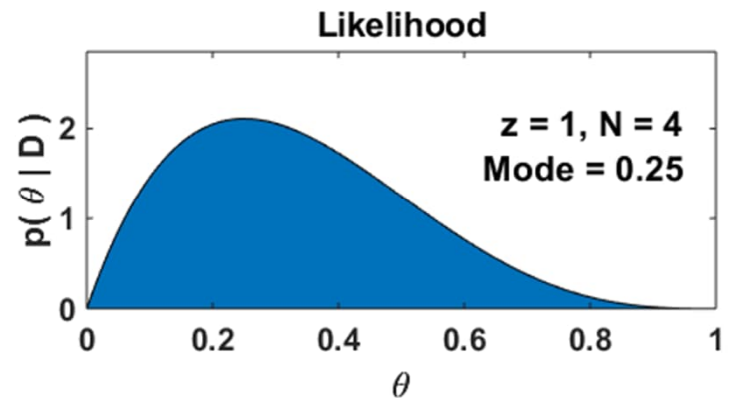
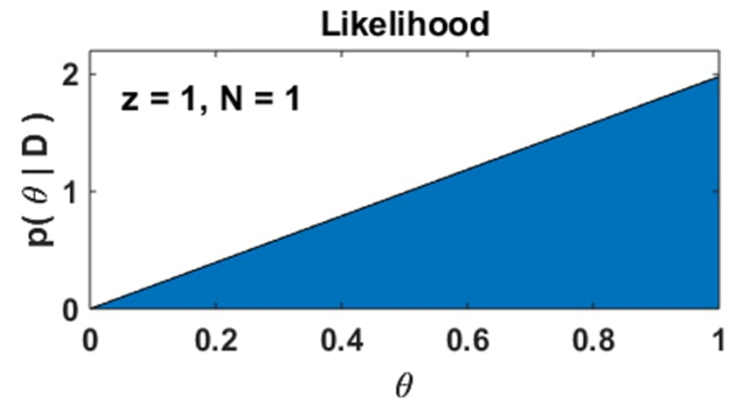
$$p(y | \theta) = \theta^y (1 - \theta)^{1-y}$$



Multiple coin flips

$$p(y|\theta) = \theta^y (1-\theta)^{1-y}$$

$$p(\{y_i\}|\theta) = \theta^{\sum_i y_i} (1-\theta)^{\sum_i (1-y_i)}$$
$$= \theta^z (1-\theta)^{N-z}$$



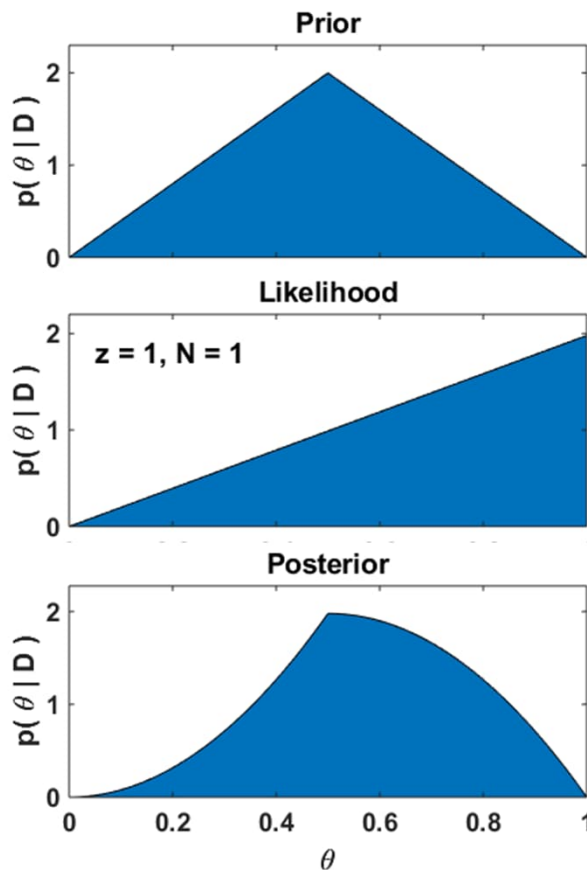
The likelihood function

- Probability of data given the parameter
- Function of the parameter, given the data

$$L(\theta) = p(y|\theta)$$

$$\int d\theta \, p(y|\theta) = \int_0^1 d\theta \, \theta^1 = \frac{\theta^2}{2} \Big|_0^1 = \frac{1}{2}$$

How will Bayes work?

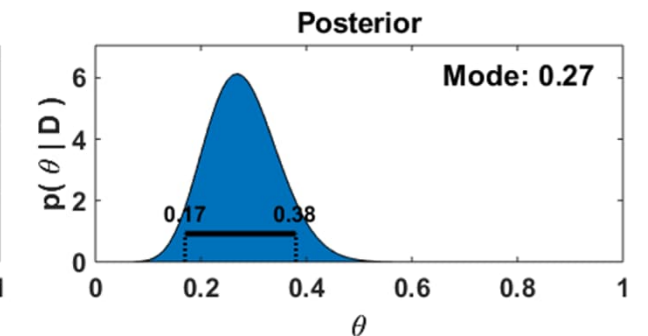
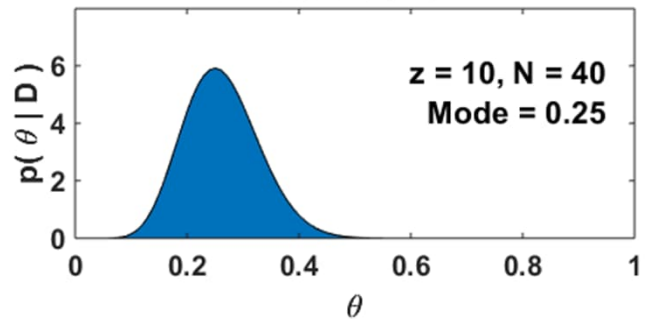
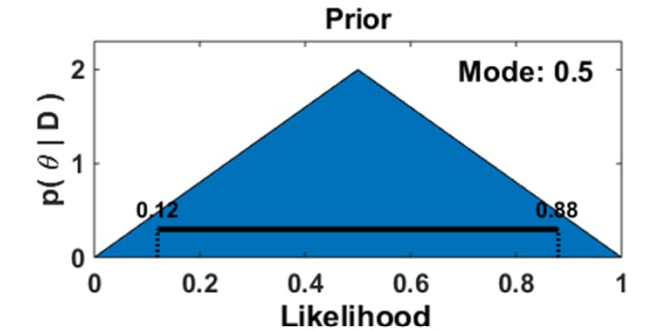
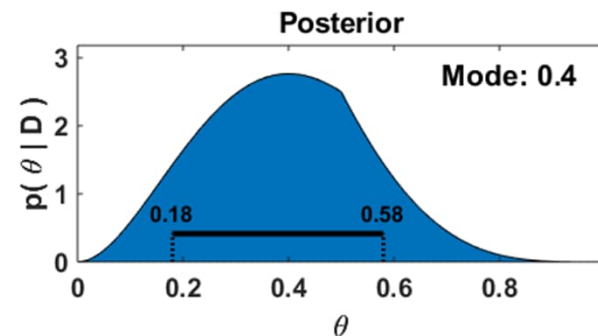
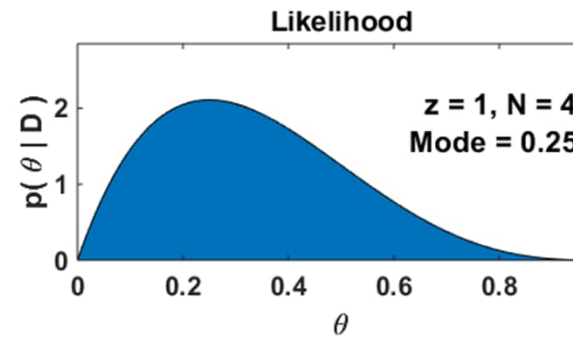
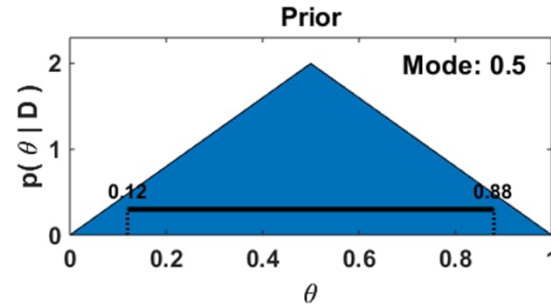


$$p(\theta | D) = \frac{p(D | \theta) p(\theta)}{p(D)}$$

- Essentially, we simply multiply prior and likelihood
- To get posterior

Multiple coin flips

- As we add more data, the data will dominate the prior



The prior is a probability distribution

- Must integrate to 1

$$\int d\theta \, p(\{y_i\}|\theta) = \int_0^1 d\theta \, \theta^z (1-\theta)^{N-z} \neq 1$$

z : Number of 1s

N : Number of attempts

$$\int d\theta \, p(\theta) = 1$$

Define the Beta distribution

- A probability distribution on θ

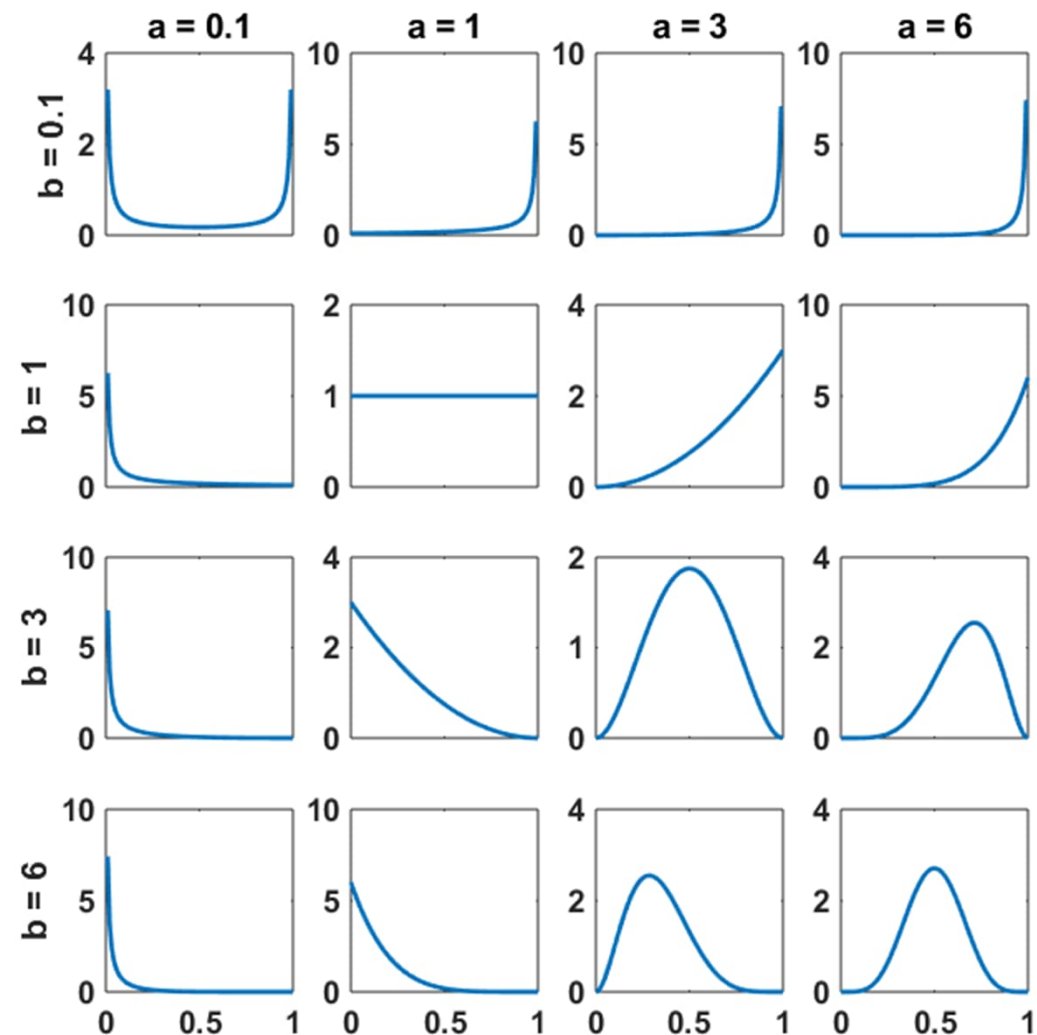
$$B(a, b) = \int_0^1 d\theta \theta^{a-1} (1-\theta)^{b-1}$$

$$\mathbf{B}(a, b) = \frac{\theta^{a-1} (1-\theta)^{b-1}}{B(a, b)}$$

$$\int_0^1 d\theta \mathbf{B}(a, b) = \frac{B(a, b)}{B(a, b)} = 1$$

What does the Beta distribution look like?

- A large variety of possible types of “plausible” probabilities
- This might be a good family of priors!



What happens if you choose a beta prior?

- We can use Bayes to find the posterior

$$p(\theta) = \mathbf{B}(a, b) = \frac{\theta^{a-1} (1-\theta)^{b-1}}{B(a, b)}$$

$$p(\{y_i\}|\theta) = \theta^z (1-\theta)^{N-z}$$

$$p(\theta|\{y_i\}) = \frac{p(\{y_i\}|\theta) p(\theta)}{p(\{y_i\})} = \frac{\theta^z (1-\theta)^{N-z} \theta^{a-1} (1-\theta)^{b-1}}{p(\{y_i\}) B(a, b)} = \frac{\theta^{z+a-1} (1-\theta)^{(N-z)+b-1}}{p(\{y_i\}) B(a, b)}$$

What happens if you choose a beta prior?

- We can use Bayes to find the posterior

$$p(\theta) = \mathbf{B}(a, b) = \frac{\theta^{a-1} (1-\theta)^{b-1}}{B(a, b)}$$

$$p(\{y_i\}|\theta) = \theta^z (1-\theta)^{N-z}$$

$$\begin{aligned} p(\theta|y) &= \frac{\theta^{z+a-1} (1-\theta)^{(N-z)+b-1}}{p(\{y_i\}) B(a, b)} = \frac{B(z+a, (N-z)+b) \mathbf{B}(z+a, (N-z)+b)}{p(\{y_i\}) B(a, b)} \\ &= \mathbf{B}(z+a, (N-z)+b) \end{aligned}$$

Posterior is compromise of prior and data

$$\text{Posterior mean} = \frac{z + a}{z + a + (N - z) + b} = \frac{z + a}{N + a + b}$$

$$= \frac{z}{N + a + b} + \frac{a}{N + a + b}$$

$$= \frac{\frac{z}{N}}{\frac{N}{N + \kappa}} + \frac{\frac{a}{a + b}}{\frac{\kappa}{N + \kappa}}$$

Data mean

Prior mean

Posterior is compromise of prior and data

$$\text{Posterior mean} = \frac{z + a}{z + a + (N - z) + b} = \frac{z + a}{N + a + b}$$

$$= \frac{z}{N + a + b} + \frac{a}{N + a + b}$$

$$= \frac{z}{N} \frac{N}{N + \kappa} + \frac{a}{a + b} \frac{\kappa}{N + \kappa}$$

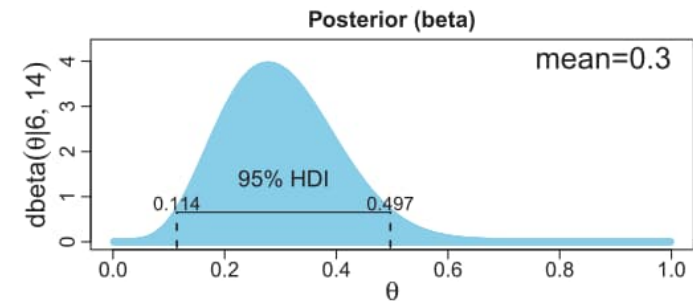
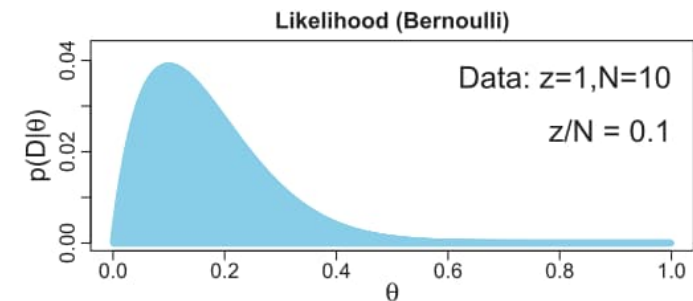
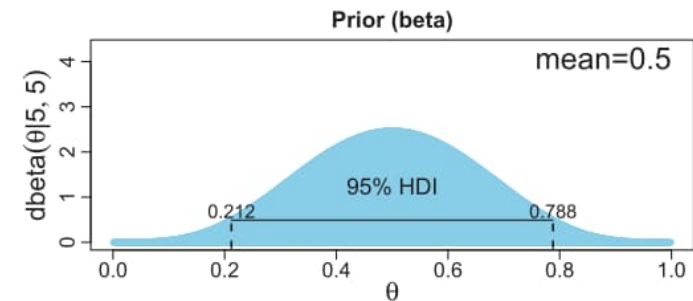
Data weight

Prior weight

Showing how posterior is a compromise

- Prior has concentration of 10
- Likelihood is result of 10 flips
- Posterior mean is right in the middle

$$\text{Posterior mean} = \frac{z}{N} \frac{N}{N + \kappa} + \frac{a}{a + b} \frac{\kappa}{N + \kappa}$$



Part 7

- Bayes' rule: Gaussian noise

Normal likelihood

For a N data points, x_i , all with known “noise” σ_x

$$\text{Likelihood: } p(\{x_i\} | \mu, \sigma_x) = \left(\frac{1}{\sqrt{2\pi}\sigma_x} \right)^N e^{-\frac{\sum_i (x_i - \mu)^2}{2\sigma_x^2}} \quad \bar{x} = \frac{1}{N} \sum_i x_i$$
$$\sigma_{\bar{x}}^2 = \frac{\sigma_x^2}{N}$$

$$x_i | \mu, \sigma_x \sim N(\mu, \sigma_x)$$

$$\text{Prior: } \mu \sim N(\mu_0, \sigma_0)$$

$$\mu_N = \frac{\frac{1}{\sigma_{\bar{x}}^2} \bar{x} + \frac{1}{\sigma_0^2} \mu_0}{\frac{1}{\sigma_{\bar{x}}^2} + \frac{1}{\sigma_0^2}}$$

$$\text{Posterior: } \mu \sim N(\mu_N, \sigma_N)$$

$$\sigma_N^2 = \frac{1}{\frac{1}{\sigma_{\bar{x}}^2} + \frac{1}{\sigma_0^2}}$$

Exercise

- You see the box and it seems like it will be light. Say $\mu_0 = 3.5\text{N}$
- You aren't sure though. You think your guess is good to $\sigma_0 = 1.0\text{N}$
- When you lift the box, your perception is that it is actually heavier. It seems to be $x = 6\text{N}$
- But, of course, your perception is noisy. $\sigma_x = 1.0\text{N}$
- If your brain is doing optimal estimation, what is your new estimate of the weight of the box?
- You continue to lift the box 4 more times and perceive the weight to be: 6, 7, 7, 4, 5. What is your final estimate of the weight and how much noise is there in the estimate?